A Grid-Independent Methodology for Radiative Flux Mapping and Assessment of Cylindrical Solar Receivers

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Supplementary Information

S1. Trapezoidal integration of the ASTM G173-03 spectrum

The cumulative spectral irradiance function $G_{\text{cum}}(\lambda)$ is evaluated numerically from the discretised ASTM G173-03 tabulated spectrum $\{(\lambda_i, G_{b,i})\}$ using the trapezoidal rule,

$$G_{\text{cum}}(\lambda_i) = G_{\text{cum}}(\lambda_{i-1}) + \tfrac{1}{2}(G_{b,i} + G_{b,i-1})(\lambda_i - \lambda_{i-1}), \qquad G_{\text{cum}}(\lambda_0) = 0 \tag{1}$$

This initialises the cumulative sum at zero and accumulates trapezoidal areas between consecutive spectral points. The band clipping edges a_j and b_j for each wavelength band λ_j are

$$a_j = \max\!\left(\lambda_j - \tfrac{\Delta\lambda}{2},\ \lambda_{\min}\right), \qquad b_j = \min\!\left(\lambda_j + \tfrac{\Delta\lambda}{2},\ \lambda_{\max}\right) \tag{2}$$

The in-band irradiance follows as $\Delta G_j = G_{\text{cum}}(b_j) - G_{\text{cum}}(a_j)$.

S2. Theoretical noise scaling of the three reconstruction methods

The three methods exhibit fundamentally different noise scaling with ray count and grid resolution.

For the bin-based method, the expected number of rays per bin of area ΔA_{grid} is $\bar{n} = N_{\text{rays}}/(N_\theta N_s)$. The relative noise scales as

$$\text{Noise}\%_{\text{BB}} \propto \frac{1}{\sqrt{\bar{n}}} = \sqrt{\frac{N_\theta N_s}{N_{\text{rays}}}} \tag{3}$$

Grid refinement at fixed N_{rays} amplifies noise — the resolution-noise coupling at the root of the checkerboard problem.

For the k-NN estimator, exactly k rays contribute to every evaluation point by construction,

$$\text{Noise}\%_{\text{kNN}} \propto \frac{1}{\sqrt{k}} \tag{4}$$

This is independent of evaluation grid resolution, motivating the k-optimisation study of Section 5.2.

For the KDE estimator with Scott's bandwidth $h \propto N_{\text{rays}}^{-1/6}$, the effective sample size within the kernel footprint scales as $N_{\text{rays}} h^2$, giving

$$\text{Noise}\%_{\text{KDE}} \propto \frac{1}{\sqrt{N_{\text{rays}}\, h^2}} \propto N_{\text{rays}}^{-1/3} \tag{5}$$

Here $h = h_{\text{eff}} = h_{\text{Scott}} \cdot \sqrt{\hat{\sigma}_c \hat{\sigma}_s}$ is the geometric-mean physical bandwidth computed after inverse normalization.

This is both grid-independent and self-adjusting: the noise floor contracts automatically as more rays are simulated, without any user intervention.

S3. FFTKDE validation against scipy gaussian_kde

Table S1 reports the percentage difference in peak flux between FFTKDE and the direct-summation scipy reference implementation at matched N_λ and grid resolution. Positive values indicate FFTKDE exceeds scipy. Mean flux agreement between the two implementations is within 0.01% in all cases.

Table 1: Percentage difference in peak flux: FFTKDE relative to `scipy.stats.gaussian_kde`, at $N_\lambda \in [500, 10000]$ rays per wavelength band. All KDE results for $N_\lambda > 10000$ in the main study use FFTKDE exclusively.

Grid	$N_\lambda = 500$	1000	2000	5000	10000
20×20	0.87	0.02	0.68	1.50	0.63
80×80	1.98	2.77	1.32	1.45	1.21
300×300	1.96	2.47	1.93	1.71	1.23
60×400	2.20	2.58	1.64	1.78	1.47
80×600	1.99	2.80	1.98	1.81	1.54
100×1000	1.95	2.57	1.93	2.15	1.64

Maximum deviation: 2.80% (80×600, $N_\lambda = 1000$). Mean absolute difference: 1.6%.

The deviation arises because FFTKDE evaluates the continuous Gaussian density on an internal 1024×1024 grid before interpolating to the export resolution. This finer internal grid captures the density maximum more precisely than direct evaluation at export grid centroids, making FFTKDE a marginally more accurate estimate of the true continuous peak. The difference is smaller than the run-to-run Monte Carlo variability of the peak cell and does not affect any FTP evaluation in the main study.

S4. FTP composite score sensitivity to component weighting

Figure S1 shows the FTP composite score for all matched-resolution cases under three weight schemes: W1 (equal, $w_i = 0.20$), W2 (SSI-dominant, $w_{\text{SSI}} = 0.50$), and W3 (accuracy-dominant, $w_{\text{ETA}} + w_{\text{PTA}} = 0.60$). The method ranking BB $<$ k-NN $\approx$ KDE is preserved under all three schemes. BB deficits are distributed simultaneously across PTA, SSI, and RCR; no single-metric reweighting eliminates all three.

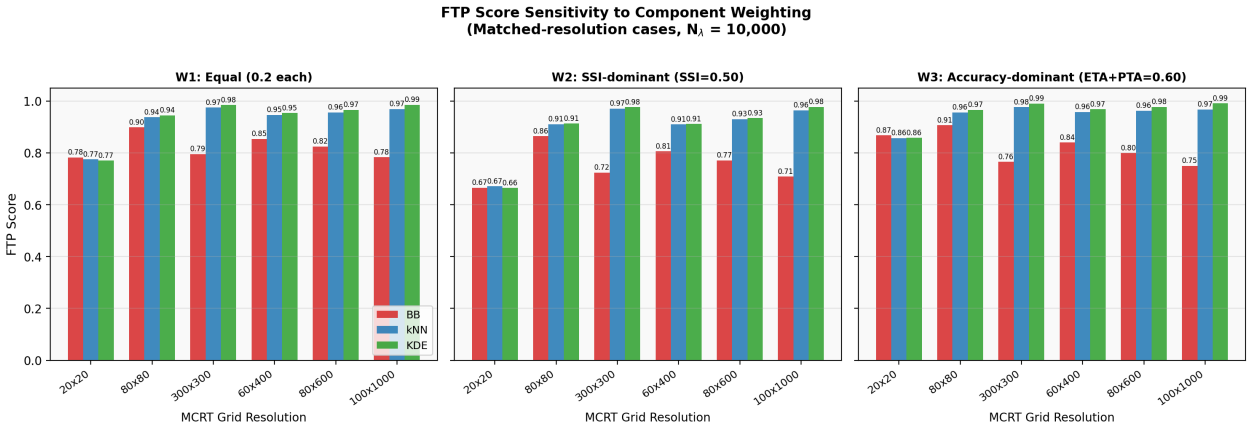


Figure 1: FTP composite score for matched-resolution cases at $N_\lambda = 10000$ under three weight schemes. Red: BB; blue: k-NN; green: KDE. The BB/(k-NN, KDE) ranking is stable across all schemes.